higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CONCRETE STRUCTURES
NQF LEVEL 4

22 NOVEMBER 2019

This marking guideline consists of 7 pages.

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QUESTION 1

1.1
1.1.1 B
1.1.2 B
1.1.3 C
1.1.4 D
1.1.5 C

(5 × 1) (5)

1.2
- Concrete cover
- Concrete mix
- Compacting
- Curing (4)

1.3 Hydration is the chemical reaction between the cement and water,√ a process that occurs during the setting and hardening of concrete.√ (2 × ½) (1)

[10]

QUESTION 2

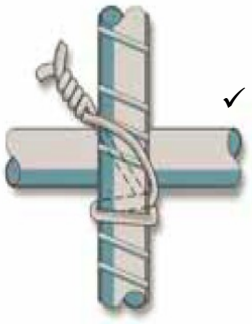
2.1
- The type of ties used and their position must be such that the required finish is achieved.
- Ties and spacers left in position must not spoil the desired appearance or affect the durability of the structure, for example by causing spalling or rust staining or by allowing the passage of moisture.
- The ends of the wire must always be neatly cut off and bent back. This is to make sure that the embedded portion does not extend into the specified minimum cover from the formed surface of the concrete.
- Ties cast into spacer blocks should not extend deeper into the spacer blocks by more than half the depth of the spacer blocks. The concrete cover over the projecting ends of ties or clamps must comply with specified concrete cover requirements.
- Reinforcement, including stirrups, links and tying wire, must not encroach into the specified cover. (Any 5 × 1) (5)

2.2



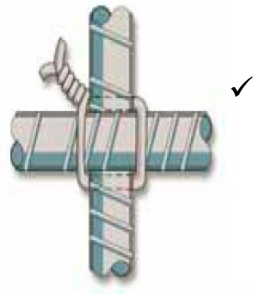
Slash tie ✓

Slash ties are the ties that are used most often. They are used to tie together the areas between main ties.✓ Slash ties are used in a slab on the bottom and top mat.✓



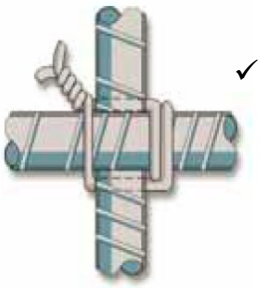
Ring slash tie✓

You can add another turn around one of the bars in a slash tie. The extra turn or loop around the bar is called a ring. With this extra ring, the tie prevents a horizontal bar from slipping down✓ and is good for tying reinforcing steel in walls.✓



Hairpin tie✓

Hairpin ties are the strongest ties and tie main bars together very securely so that they cannot move.✓ Hairpin ties secure the cage completely before the concrete is placed in the formwork.✓



Ring hairpin tie✓

A ring Hairpin tie is the same as the hairpin tie but you add another turn around one of the bars in a hairpin tie.✓ The extra ring prevents slipping and is ideal to tie reinforcing steel in walls.✓



Crown tie✓

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Splice tie✓

Splice ties are used to tie lap splices together where two bars overlap.✓ A double wire is used. The wire is first folded double and then this double length is used as though it was a single piece of wire.✓

(Any 5 × 4) (20)

2.3 2.3.1 Slash tie

- Step 1: Wrap the wire diagonally once around the two crossing bars with the two ends on top.
- Step 2: Tighten and twist the ends of the wire together.

2.3.2 Hairpin tie

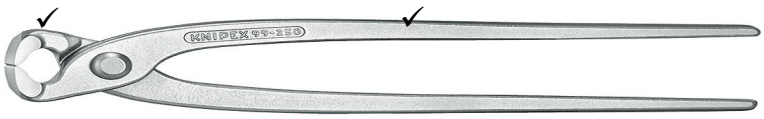
- Step 1: Wrap the wire halfway around one of the bars on either side of the crossing.
- Step 2: Tighten the wire squarely or diagonally around the crossing bat and twist the ends together.

2.3.3 Splice tie

- Step 1: Fold the wire double and tighten around the overlapping bar (lap splices).
- Step 2: Tighten and twist the ends of the wire together.

(3 × 2) (6)

2.4 2.4.1



Tying wire nips

2.4 2.4.1



Bar wire twister

(2 × 2) (4)
[35]

QUESTION 3

3

✓ ✓ ✓ ✓

Item	Inspection	✓/x	Action
1	The correct number of bars are installed		
2	The correct type of bars are installed		
3	The correct diameter bars are installed		
4	Bars are clean and oil free		
5	Bars are all in the correct position		
6	Bars are properly spaced		
7	Adequate access between bars to insert the poker vibrator for compacting		
8	Twisted ends of ties are bent inwards		
9	All laps are of the correct length		
10	All top steel are adequately supported by stools or chairs		
11	Cages or mats are rigid/strong so that they will not move during placing		✓

Inspection checklist (5 × 1)
Inspection checks (10 × ½) **[10]**

QUESTION 4

4.1
- Concrete must be placed within an hour after being discharged
- The concrete should be covered during transportation
- Concrete should be placed as near as possible to its final position
- Concrete should not be deposited in heaps
- Concrete must be deposited in horizontal layers
- The placing of concrete must be continuously to avoid cold joints

(Any applicable 5 × 1) (5)

4.2
- Segregation during pour.
- Leakage between the kicker and the bottom of the formwork.
- The quality of the concrete isolated pockets of honeycombing can occur when a stiff mixture is being used without sufficient consolidation. (3)

4.3
- Paste leaking through form faces joints.
- Bulging of the form face or any of the members supporting the formwork.
- Movement or buckling of false work or any of the members.
- Creaking or groaning sound from any part of the formwork.
- The stretching or snapping of ties in walls and columns.
- The crushing of timber components against a restraining steel wall tie washers against timber walers or soldiers. (Any applicable 5 × 1) (5)

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4.4
- Test result A is a true slump✓ with a maximum slump limit of 120 mm and a minimum of 50 mm. With the true slump, the concrete simply subsides and keeping more or less its shape.✓ The mix design is perfect for the specific task and don't have to be modified.✓
- Test result B is a zero or no slump✓ at all with no deformation that is an indication of a very dry mix and a low workability.✓ More water must be added to the concrete mix after the mix ratios were checked.✓
- Test result C is a collapsed slump✓ and is an indication of a too wet mix. The Concrete collapse completely.✓ The mix design is not correct and need to be modified by reducing the water content but not changing the aggregate ratios as prescribed.✓
- Test result D is a shear slump✓ where the top portion of the concrete shear off and slips sideways.✓ Reason for a shear slump can be different reasons e.g. the compacting was not the same for all layers, the water content is too high for the specific mix ratio or the mix lacks cohesion. In the case of a shear slump it is recommended to perform the test again and then if the result is the same, the water content or aggregate properties should be adjusted e.g. finer sand or smaller stone.✓

(Any 12 × 1) (12)
[25]

QUESTION 5

5.1.1 Hot – 7 days

5.1.2 Cool – 9.5 days

5.1.3 Normal – 7 days

(3 × 2) (6)

5.2
- Has the engineer/supervisor given the go-ahead?
- Are the instructions clear?
- When was this concrete cast?
- Are there specific instructions from the design professional as to procedure?
- Has the weather been cold?
- Has the prescribed minimum period elapsed?
- Is there a curing procedure?
- Has there been below zero temperature soon after casting?
- Do you have your hard hat and safety glasses with you?
- Is there a place to stack the formwork neatly?
- Are you sure that this is the formwork that must be struck now?

(Any 4 × 1) (4)
[10]

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QUESTION 6

6.1 6.1.1 You should store timber panels horizontal on a level base where the weight of the above panels helps the lower panels to stay flat.✓ This helps to reduce warping.✓ You should also cover the panels with water proof sheeting that will reduce evaporation of moist and that will slow down the drying-out process.✓ (Any applicable 3 × 1) (3)

6.1.2 They must be stored undercover in a shed or container.✓ Store them face-to-face and back-to-back to prevent the back from scratching the face.✓ Always lightly oil them before storing to avoid corrosion.✓

(Any applicable 3 × 1) (3)

6.2
- Fixed and free-standing shelving
- Lockers
- Bins
- Drawers

(4)
[10]

TOTAL: 100

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